You (probably) don't understand Claude Code memory.

A practical guide to global, project, and rule-level memory design

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This week, I walked my team through how I have been using Claude Code in real projects.

The discussion quickly drifted away from prompts and into something more interesting: why Claude sometimes feels incredibly reliable, and other times strangely inconsistent. Almost every example traced back to the same root cause — a misunderstanding of memory.

That conversation stuck with me. It felt like something worth sharing more broadly, because Claude Code memory is powerful, but only if you stop thinking of it as “memory” in the human sense. This post is my attempt to explain the mental model that finally made it click for me. I break down how Claude Code memory actually

works, how its different layers interact, and how to design memory files that make Claude a predictable, reliable collaborator in real projects.

If you are interested, in learning how to make the most out of Claude's memory, this post is for you.

What this blog will cover

- **What is Claude Code memory, really?**
  What "memory" means in practice, and why it is less about tone or style and more about how Claude decides when to plan, ask questions, or act.
- **How do memory hierarchies work across global, team, and project levels?**
  A clear mental model of how different layers of memory interact, with concrete examples from data science and data engineering workflows.
- **What belongs in a global `CLAUDE.md` file? (and my personal example)**
  How to define stable behavioural rules that apply everywhere and why treating this file as "Claude's personality" is a mistake.
- **How should project-level `CLAUDE.md` files be structured?**
  Practical guidance on organising rules by themes and constraints, without over-describing the project or fighting the codebase.

- **What does this look like in real projects?**
  Two worked examples: one for building an ML model, and one for running an ETL pipeline, showing how memory choices directly change Claude's behaviour.
- **How should these memory files evolve over time?**
  When to update them, when to leave them alone, and when clearing memory is the healthiest option.

Let's get started!

What is Claude Code memory, really?

At its simplest, **Claude Code memory is just text**.

More precisely, it is a set of Markdown files that Claude reads every time a new session starts. These files contain instructions that are injected into Claude's context before it ever sees your first message.

That simplicity is important. There is no magic database, no long-term learning, and no mysterious internal state. Claude (and probably all LLM systems) does not "remember" in a human sense.

It **re-reads instructions**, every time. Something like this 👇

```
New Claude Code session starts
           ↓
Load Markdown memory files
(global / team / project)
           ↓
Inject contents into prompt context
           ↓
Claude reasons over:
- goals
- constraints
- allowed actions
           ↓
User sends first message
           ↓
Claude responds (with memory applied)
```

Why Markdown matters here

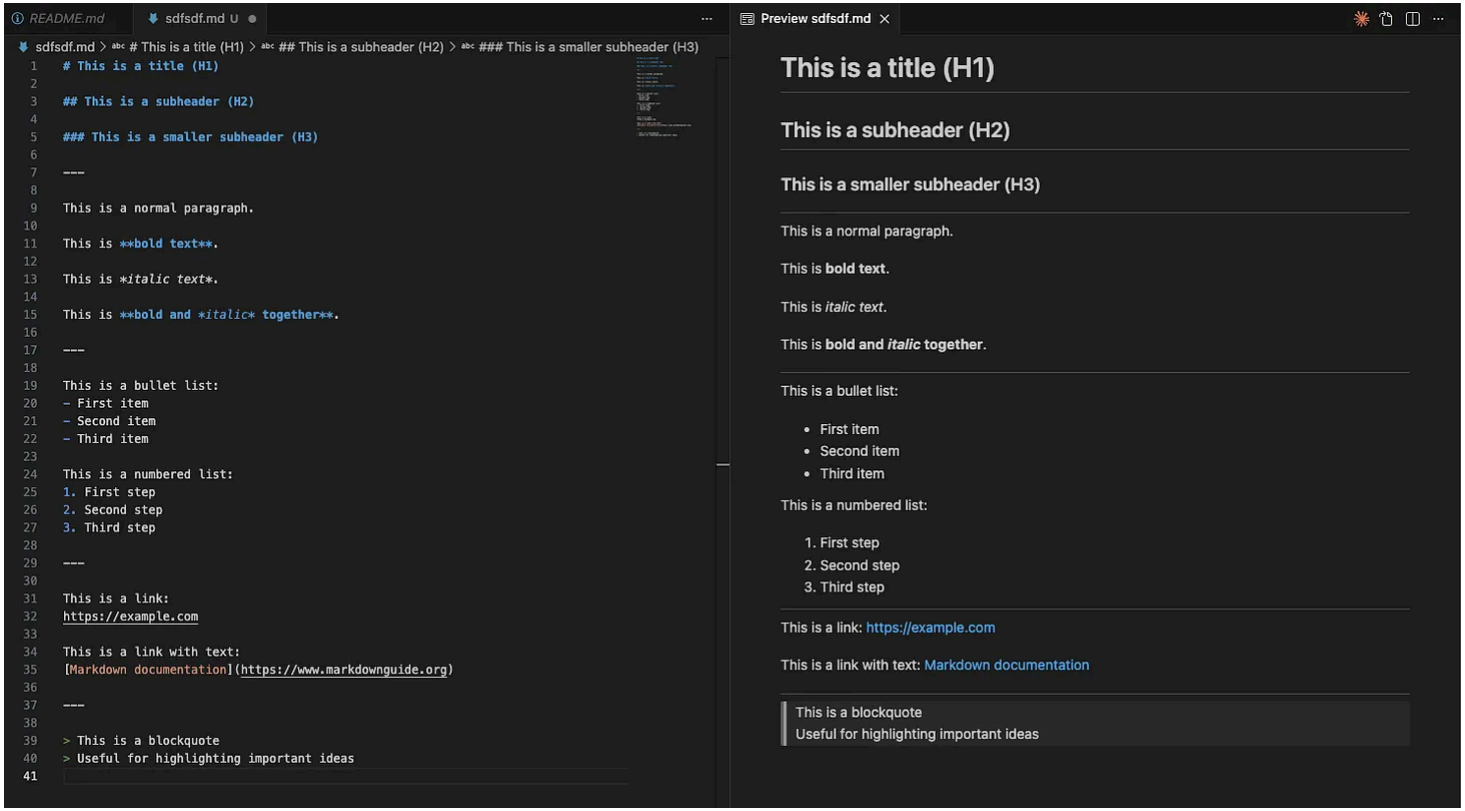
Claude Code memory lives in **Markdown files**, typically named `CLAUDE.md`. That choice is not accidental.

Markdown is a format that large language models understand extremely well:

- It has clear hierarchy through headers and subheaders

- It supports lists, bold, and *italics* to signal emphasis
- It separates concerns naturally without verbosity

In other words, Markdown already encodes *structure*, and LLMs are very good at exploiting structure. This also makes the files pleasant for humans. You can read them as plain text, edit them in any IDE, and preview their rendered form side by side.



How memory is actually injected

At a high level, Claude Code memory works like this:

- Claude starts a new session
- The relevant memory files are read
- Their contents are **inserted into the prompt context**
- Only then does Claude process your request

In practical terms, Claude memory is *very close* to a copy-and-paste operation. This is where many people draw the wrong conclusion. They think:

> "If this is just text being injected, I should describe everything. Every preference. Every edge case. Every possible interaction."

That is wrong a lot of reasons which we will describe later. But the effects I have seen is that people start writing these instructions to try to give Claude "personality".

Personality is, simply, the wrong metaphor. Now, this instinct is understandable because when people hear "memory", they think about tone, voice, friendliness, verbosity. They try to encode how Claude should *sound*.

But, Claude Code memory is not primarily about conversation.

It is about ***coordination***.

Memory for coordination

A better framing is this:

> Claude Code memory defines how Claude behaves as a collaborator, not how it chats.

When Claude reads memory, it is not optimising for phrasing. It is deciding *how to operate*.

Questions like:

- Should I propose a plan before acting?
- How much ambiguity is acceptable before I stop and ask?
- Am I allowed to edit files directly?
- Should I prioritise correctness or speed?
- How risky is this change?

These decisions happen ***before*** any text is generated. By the time you see an answer, the important choices have already been made.

This is why a short, well-designed memory file can feel transformative (and a long, verbose one can feel ignored).

A crucial segue: memory is not flat

One final clarification before we go deeper.

Although memory is injected as text, **it is not treated as a single flat block**. Claude is more intelligent than that. It reasons over *where* instructions come from and *how broadly they should apply*.

This is why Claude Code supports **memory hierarchies**: global, team, and project-level memory. Different files apply at different scopes, and Claude uses that structure to decide which rules matter most in a given situation.

That hierarchy — not verbosity — is what makes memory powerful.

And that is what we will tackle next.

How do memory hierarchies work across global, team, and project levels?

Claude Code memory is not a single file. It is organised as a **small hierarchy of Markdown files**, each applying at a different scope.

The hierarchy at a glance

A simple way to visualise this is to borrow a repository-style layout:

```
~/.claude/
└── CLAUDE.md                 # Global memory (always loaded)

project-repo/
├── CLAUDE.md                 # Project memory (always loaded)
├── .claude/
│   └── rules/
│       ├── testing.md        # Applied conditionally
│       ├── viz-standards.md  # Applied conditionally
│       ├── ml-experiments.md # Applied conditionally
│       └── notebooks.md      # Path-scoped via `paths:` frontmatter
├── src/
├── tests/
├── notebooks/
└── README.md
```

Conceptually, Claude sees this as:

```
Enterprise / User memory
        ↓
Global CLAUDE.md
```

```
        ↓
Project CLAUDE.md
        ↓
Project rules (.claude/rules/*.md)
        ↓
(Conditional rules, if applicable)
        ↓
Your first message
```

Are all memory files loaded every time?

When you start a session, Claude Code *automatically injects* memory files into its prompt context based on a defined hierarchy, where broader contexts (enterprise or user) are loaded first and more specific contexts (project) override them.

For example:

- The **global** `CLAUDE.md` is always applied
- The **project** `CLAUDE.md` is always applied when working inside that repository
- **Modular rules** in `.claude/rules/*.md` are also loaded and become part of the project's context. These files can be kept topic-specific (for example, data-visualisation standards or testing guidelines).

However, you can make some of these files "conditional"

- **Conditional path-scoped rules.** You *can* make rules that only apply when editing certain files by adding a `paths` frontmatter section — e.g. rules that only activate when working on Python test files or notebooks. This is the closest built-in mechanism Claude has to "only load relevant rules."
- You can point Claude to specific files using pointers like `@path/to/import` letting you organise deeper guidance externally while controlling exactly when it becomes part of the loaded context.

(Don't worry, we will cover all these details later and in different posts)

Despite Claude loading or not certain files, when it does, the contents of these files are injected into the prompt context before your first message.

This has an important implication.

Context is finite, even if memory feels free

Because memory is injected as text, every line you add consumes context budget.

This is why memory hierarchies exist in the first place:

- Global memory should stay *small and stable*
- Project memory should be *focused and scoped*

- Repetition across files is costly
- Verbosity compounds quickly across sessions

This is also why **treating memory as a place to document everything is a *mistake***. You are not writing a wiki. You are shaping behaviour under tight constraints.

We will come back to this when we discuss project-level memory patterns and techniques like referencing, grouping, and controlled expansion.

Now that we understand how Claude Code memory is structured and loaded, we can move from mechanics to intent. We will start with the **global** `CLAUDE.md` — the one file that applies everywhere and shapes how Claude behaves across all projects.

What belongs in a global `CLAUDE.md` file? (and my personal example)

The global `CLAUDE.md` file is the **highest-leverage memory file** you will write.

The goal is to define a few stable expectations that will not change from one project regardless of what these projects are (content generation, coding, researching, etc).

This is why the global `CLAUDE.md` should feel like a *working contract*. Not documentation, nor personality design.

Below you can see some high level guidelines on what to include or avoid in this global file:

What to include	What to avoid
How Claude should plan work (when to propose a plan vs act directly)	Project descriptions or repository overviews
Rules for asking clarifying questions when requirements are ambiguous	File structures or folder layouts
Your risk posture (when Claude must wait for approval)	Domain-specific logic or technical constraints
Expectations around file edits and tool usage	Coding conventions or style guides
How Claude should communicate uncertainty and trade-offs	Temporary preferences or one-off instructions
Stable collaboration norms that apply everywhere	Task-specific checklists or workflows
Rules that change rarely, if ever	Anything likely to change next month

Source: Jose Parreño • Get the data • Created with Datawrapper

My own `CLAUDE.md` file

```
# Jose's Global Claude Code Preferences

## Relationship and Voice

- In CLAUDE.md files, Jose and Claude avoid pronouns:
  - Use “Jose” (or “the human”) and “Claude”.
  - Example: “Jose writes, Claude edits.”
- In normal conversation (chat), Jose and Claude use natural language pronouns as
usual.

## Primary Objective

- Claude optimises for correctness and clarity over speed or cleverness.
- Claude prefers long-term maintainability and ease of understanding.

## Planning and Change Control

Claude proposes a brief plan before implementing non-trivial work.

Non-trivial includes:
- new functionality
- architectural or behavioural changes
- refactors or changes across multiple files
- tasks with multiple reasonable approaches
- substantial output (for example: a new file >50 lines, code spanning multiple
```

```
functions, or documentation >300 words)

Process:
- Claude proposes: (1) approach, (2) key trade-offs, (3) a short checklist of steps.
- If requirements are ambiguous, Claude asks the minimum number of clarifying questions
needed to proceed.
- For non-trivial work, Claude waits for explicit approval before implementation.

Trivial work may be done directly:
- typos, mechanical edits, obvious bug fixes, tiny refactors

## Communication Style

- Claude is direct and specific in critique and feedback.
- Claude avoids hedging language when making technical recommendations.
- Claude uses bullet points for summaries, decisions, and feedback.
- Claude explains tool usage proportionally:
  - trivial actions: minimal explanation
  - non-trivial decisions: clear reasoning

## Defaults for Outputs

- When Claude proposes a plan, Claude keeps it short (3–7 bullets).
- When Claude provides recommendations, Claude includes:
  - what Claude would do
```

```
  - why
  - what could go wrong (if relevant)
```

My `CLAUDE.md` file is:

1. ~50 lines long.
2. Defines operational behaviour and risk posture.
3. Draws a clear line between trivial and non-trivial work.
4. Makes feedback useful, not polite or flattering.

In other words, I describe how Claude should behave when collaborating with me.

How should project-level `CLAUDE.md` files be structured?

If the global `CLAUDE.md` defines *how Claude behaves everywhere*, the project-level file defines **how *a specific repository works*.**

- Anthropic's own guidance describes project memory as the place for *important project information, conventions, and frequently used commands.*

- Community practice has converged on a similar framing: A project `CLAUDE.md` is best thought of as **AI onboarding plus an operating manual** — not as a dumping ground for everything you know about the system.

What a project `CLAUDE.md` is for

A well-structured project file helps Claude answer questions like:

- What does this system do, at a high level?
- What constraints matter here?
- How is this repo organised?
- How do people typically build, test, and change things?
- What would be dangerous to modify casually?

It should contain information that materially changes Claude's decisions, not information Claude can easily infer from reading the code.

How descriptive should it be?

Brief, explicit, and high-signal. Best practice is to include a short description of the project, usually just a handful of bullets:

- what the system does,

- what matters most (constraints),
- what is explicitly out of scope,
- where the important directories live.

If you already have a good README or architecture document, reference it rather than duplicating it. Repetition wastes context and increases the chance of drift.

How long should it be?

There is no hard limit, but real-world usage shows a clear pattern:

- **20–80 lines** → small repos, libraries, single services
- **80–200 lines** → typical product services, multi-module repos
- **Beyond ~200 lines** → once a project `CLAUDE.md` starts to feel like a "long document", it becomes harder to maintain and easier to contradict. This is exactly why modular rules exist.

Architecture and plans: keep them lightweight

Project memory is a good place for a short "**architecture at a glance**" section:

- key components,
- how they relate,

- where they live in the repo.

It is *not* a good place to maintain an execution plan or a running checklist. Plans change too often, and memory files should not become fossils.

If you need repeatable workflows, use issue trackers, commands, or purpose-bound skills, not a static plan embedded in memory.

Avoiding bloat with `.claude/rules/`

The most important structural principle is this: **The root project `CLAUDE.md` should stay small and stable.**

Anything that is deep, topic-specific, path-specific, or only relevant some of the time, is usually better placed in **modular rule files** under `.claude/rules/`.

---

*The project `CLAUDE.md` becomes the **index**.*

*`.claude/rules/` becomes the **detail**.*

With this structure in mind, we can now look at what this actually looks like in practice. In the next section, we will walk through **real project examples**, starting with a data science workflow and then contrasting it with a data engineering pipeline.

What does this look like in real projects?

Example 1: a data science project building an ML model

Let us start with a data science project whose goal is to **build, evaluate, and iterate on a machine learning model**.

This is a good first example because data science projects tend to be exploratory, iterative and prone to accidental scope creep. A well-designed project-level `CLAUDE.md` helps Claude navigate that uncertainty without becoming either reckless or paralysed.

What this project needs Claude to understand

Before writing any memory, it helps to articulate the *failure modes*. In a typical modelling project, Claude can easily over-optimise code quality too early and refactor exploratory code that should stay flexible. So the project memory should focus on **guardrails**, not explanations.

```
# Project: Customer Churn Prediction Model

## What this project is
- Purpose: train and evaluate models to predict customer churn
- Primary output: offline evaluation results and model comparison
- This is an experimental modelling project, not a production system

## Key constraints
- Evaluation metrics must be comparable across experiments
- Feature leakage must be avoided
- Reproducibility matters more than raw performance

## Repo layout (high level)
- data/: raw and processed datasets
- features/: feature engineering logic
- models/: model definitions and training code
- evaluation/: metrics, validation, and analysis
- notebooks/: exploratory analysis only

## How to run experiments
- Training scripts live under models/
- Evaluation is run via evaluation/run_evaluation.py
- Do not modify evaluation logic without explicit discussion
- All experiments must log metrics and config to <tool or file>
```

```
## Modelling and experimentation rules
- Treat notebooks as exploratory; final logic should move to Python modules
- New features must be documented briefly in features/README.md
- Always report baseline comparisons when introducing a new model. Baseline is
logistic_regression_v1 located in [directory]

## Non-goals
- This project does not handle model serving or deployment
- Performance optimisations are secondary to correctness and clarity

## Where deeper rules live
- Feature engineering standards: .claude/rules/features.md
- Evaluation conventions: .claude/rules/evaluation.md

## Environment and dependencies
- Python: <version>
- Dependencies managed via: <poetry/uv/conda/pip-tools>
- If a new dependency is required, Claude explains why and updates lock files
```

Why this works for a data science workflow

A few things are worth calling out.

- **It sets expectations about maturity.** By stating explicitly that this is *not* a production system, Claude is less likely to over-engineer early iterations.

- **It protects the evaluation layer.** Calling out evaluation logic as sensitive changes Claude's risk calculus immediately.
- **It respects exploratory work without enshrining it.** Notebooks are allowed, but they are not treated as final artefacts. Claude gets clear guidance on when code should "graduate".
- **It offloads depth into modular rules.** Detailed feature or evaluation standards are not inlined. They are referenced. This keeps the root file readable and stable.
- **What is deliberately missing.** Detailed feature definitions, long architectural explanations, experiment-by-experiment history, or a running task list. All of those would either change too frequently or belong somewhere else.

The role of this file is not to explain the project exhaustively.

It is to make Claude *harder to mislead by accident*.

Example 2: a data engineering project building an ETL pipeline

Now let us look at a very different kind of repository: a **data engineering project responsible for an ETL pipeline**. Compared to modelling work, ETL systems are less exploratory, far more stateful and much more sensitive to seemingly small changes.

What this project needs Claude to understand

In an ETL pipeline, Claude can easily:

- introduce non-idempotent logic,
- break downstream consumers by changing schemas,
- refactor code that encodes operational knowledge,
- or “simplify” behaviour that exists for historical reasons.

So the project memory must prioritise **safety, stability, and invariants**.

This is not a place for cleverness.

```
# Project: Daily Revenue ETL Pipeline

## What this project is
- Purpose: ingest, transform, and publish daily revenue data
- Primary output: curated tables consumed by reporting and downstream systems
- This is a production data pipeline

## Key constraints
- Pipelines must be idempotent (per processing date)
- Schema changes are breaking changes
- Historical backfills require explicit approval
```

```
## Repo layout (high level)
- extract/: source-specific ingestion logic
- transform/: business logic and transformations
- load/: publishing and storage
- schemas/: table definitions and data contracts (source of truth)
- tests/: data quality and pipeline tests

## How to run the pipeline
- Local execution via `make run-local`
- Production runs are orchestrated externally
- Never bypass orchestration logic for production data

## Data and schema rules
- All schema changes must be reviewed explicitly
- Backward compatibility is required unless stated otherwise
- Prefer additive changes over destructive ones
- Do not infer schemas implicitly from data or code

## Operational safety rules
- Do not introduce non-deterministic behaviour
- Idempotency must be preserved for the defined unit
- Do not change retry, failure, or publishing semantics lightly
- On failure, prefer failing the run over partial publication
- Emit clear logs and basic run metrics (row counts, sanity checks)
```

```
## State and storage
- Pipeline state is persisted only in approved storage locations
- Do not introduce new state or side tables without discussion

## Non-goals
- This project does not perform analytics or modelling
- Ad-hoc queries do not belong here

## Where deeper rules live
- Testing and data quality standards: .claude/rules/testing.md
- Schema evolution rules: .claude/rules/schema.md
- Backfill procedures and safeguards: .claude/rules/backfills.md
```

Why this works for a data engineering workflow

The contrast with the data science example is intentional.

- **It frames the project as production-critical.** That single statement changes how Claude approaches almost every decision.
- **It elevates invariants to first-class concepts.** Idempotency, schema stability, and backfills are behavioural constraints.
- **It encodes operational caution explicitly.** Claude is told where *not* to optimise and where to stop and ask.

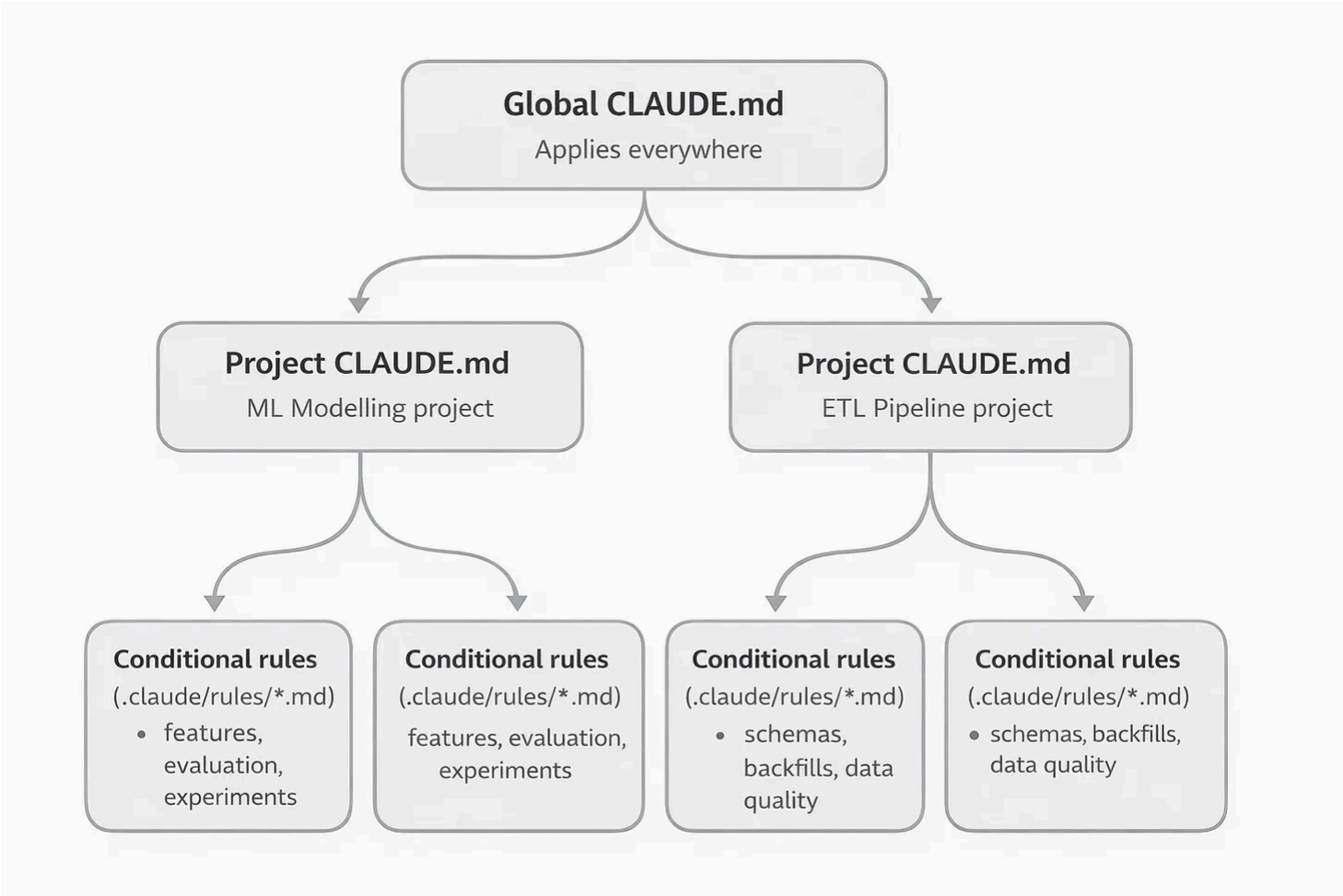
- **It pushes complexity into modular rules.** Backfills, schema evolution, and testing are deep topics. Keeping them in `.claude/rules/` avoids bloating the root file while still making guidance available when relevant.
- **What is deliberately missing.** No step-by-step pipeline walkthrough, no historical incident log, no checklist of past migrations, no execution plan. Those artefacts change too often and are better handled by runbooks, tickets, or documentation.

The purpose of this file is not to teach Claude how the pipeline works line by line. It is to prevent Claude from making **dangerous assumptions**.

The key takeaway from both examples

Both projects use a project-level `CLAUDE.md`. Both files are similar in size and structure. But they encode *very different risk profiles*.

That is the real value of project-level memory: it lets the same assistant behave appropriately in fundamentally different systems — without rewriting your global rules or fighting the tool.



How should these memory files evolve over time?

The short answer is: much less often than you think.

Both global and project-level memory files are meant to be **slow-moving**. If you find yourself editing them every week, that is usually a signal that too much is living in memory.

The examples we just saw — both the modelling project and the ETL pipeline — are intentionally written in a way that should survive months of iteration without change. You can run dozens of experiments, refactor large parts of the codebase, or add new data sources without touching project memory at all.

When you *should* update memory

Updating memory makes sense when something **fundamentally changes how Claude should behave**. There are not a lot of triggers of this, here are 2 good situations though:

1. **A repeated correction keeps coming up.** If you find yourself telling Claude the same thing three or four times, it probably belongs in memory.
2. **A risk profile changes.** Experimental code becomes production-critical, or vice versa.

Additions, not accretion

When memory does evolve, it should usually evolve by **addition and refactoring**, not by accumulation. Good patterns include:

- Moving new detail into `.claude/rules/` rather than expanding the root file
- Splitting a growing section into a focused rule file
- Removing rules that no longer influence behaviour

If a memory file only ever grows, it will eventually become harmful (remember token bloat is a real problem).

Closing thoughts

Claude Code memory is easy to misunderstand because many people think that a machine can simply retain much more memory than a human. This mean, chucking everything that comes to mind to the “memory files”.

The key idea to take away is this: **memory is not about making Claude sound better. It is about making Claude behave predictably**.

When memory is treated as a working contract rather than a personality description, a few things happen:

- behaviour becomes consistent across sessions,
- project context stops leaking into unrelated work,

- and you spend far less time correcting the same mistakes.

You do not need clever prompts or long instructions. You need small, stable rules placed at the right level.

Once you internalise that, Claude Code memory stops feeling like a mysterious feature and starts feeling like what it really is: **system design in miniature**.

Now, I want to hear from you!

In this post, we unpacked Claude Code memory from first principles — not as a mysterious feature, but as a small system you can design intentionally. But now I would love to hear from you:

- Have you already experimented with `CLAUDE.md` files in your own projects?
- Did this mental model change how you think about “memory” in LLM-based tools?
- Have you run into cases where memory helped a lot — or quietly made things worse?

Drop your thoughts, questions, or war stories in the comments. I am genuinely curious how others are using Claude Code memory in practice 👇

Further reading

If you are interested in more content, here is an article capturing all my written blogs!

All my articles, one convenient place

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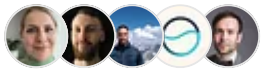

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Very well written! Saving it to read it very soon.

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